# VISIT NEARBY CEMENT INDUSTRY IF POSSIBLE OR CONSULT E-MEDIA AND LIST OUT THE RAW MATERIALS FOR CEMENT PRODUCTION AND STEPS OF MANUFACTURING OF OPC AND PPC CEMENT. ALSO STUDY THE SCOPE AND CHALLENGES OF CEMENT INDUSTRY IN NEPAL

**PROJECT WORK SUBMITTED TO DEPARTMENT OF CHEMISTRY**


**TRITON INTERNATIONAL COLLEGE**

**KATHMANDU, NEPAL**

**By**

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**Class: 12**

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**Date: 2nd February 2026**

# CERTIFICATE OF APPROVAL

It is certified that Mr. Aashish mahato has carried out the project work entitled **"VISIT NEARBY CEMENT INDUSTRY IF POSSIBLE OR CONSULT E-MEDIA AND LIST OUT THE RAW MATERIALS FOR CEMENT PRODUCTION AND STEPS OF MANUFACTURING OF OPC AND PPC CEMENT. ALSO STUDY THE SCOPE AND CHALLENGES OF CEMENT INDUSTRY IN NEPAL."** under my supervision and guidance. I recommend this project work for the requirement of secondary level degree of science by NEB.

**.....**

**Supervisor**

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# ABBREVIATION

- Ordinary Portland Cement (OPC)
- Portland Pozzolana Cement (PPC)
- Limestone ($CaCO_3$)
- Silicon dioxide ($SiO_2$)
- Aluminium oxide ($Al_2O_3$)
- Iron oxide ($Fe_2O_3$)
- Calcium oxide (CaO)
- Tricalcium silicate ($C_3S$)
- Dicalcium silicate ($C_2S$)
- Tricalcium aluminate ($C_3A$)
- Tetracalciumi aluminoferrite ($C_4AF$)
- Fly ash (FA)

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